

Truth, beauty, and data

Session 1

ISS Data Visualization with R
Faculty of International Social Sciences, Gakushuin University
Spring 2026

Plan for today

Facts, truth, and beauty

Data, truth, and beauty

Beautiful visualizations

Class details

Facts, truth, and beauty

What is truth?

Core principles of the universe?

Underlying trends in society?

Something transcendental?

Reality?

How do we find truth?

Science!



Neil deGrasse Tyson 
@neiltyson



The good thing about Science is that it's true whether or not you believe in it.

10:41 AM · Jun 14, 2013 · [TweetDeck](#)

14.3K Retweets **8.3K** Likes



But wait!

Beware of scientism!

"... promotion of science as the best or only objective means by which society should determine normative and epistemological values"

Science is not the only way

Art

Music

Literature

Religion

Nature

Nothing here is factual...

...but it all reveals truth



Cosette



King Lear



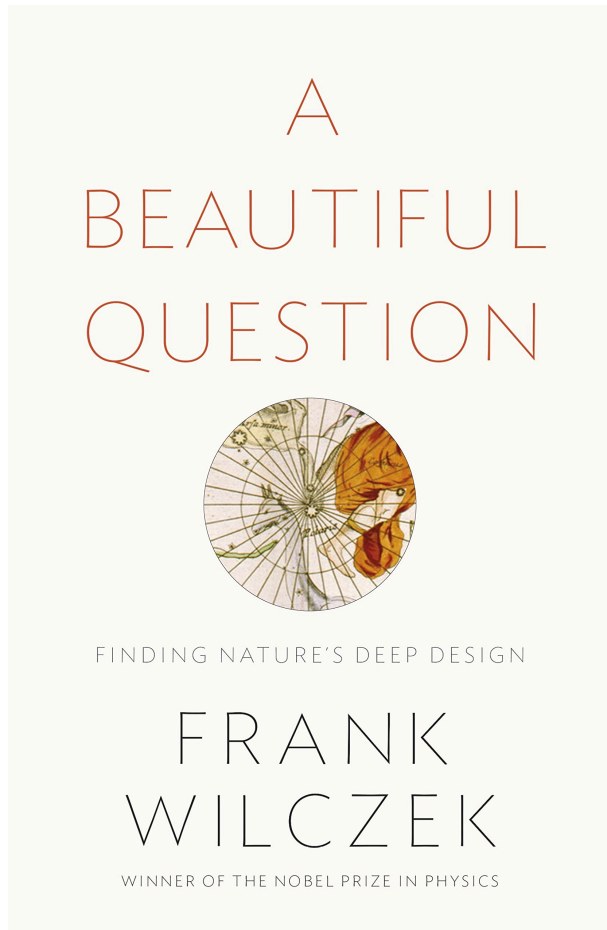
Beethoven's 9th symphony

Facts $\neq$ truth

**Where does truth come when
there are no facts?**

Beauty

Beauty in science



**This is also true for
science and math
and other more
factual realms**

Rhetoric and beauty

λόγος • λέξις

Logos • Lexis

Res • Verba

Essence • Structure

Content • Form

Truth • Beauty

Content + form

**Art is how we translate core,
essential content (or truth!)
to different forms
for specific audiences.**

Truth is beautiful

Truth $\neq$ facts

**Truth comes from aesthetic
combination of content and form**

Facts require beauty to be true

Data, truth, and beauty

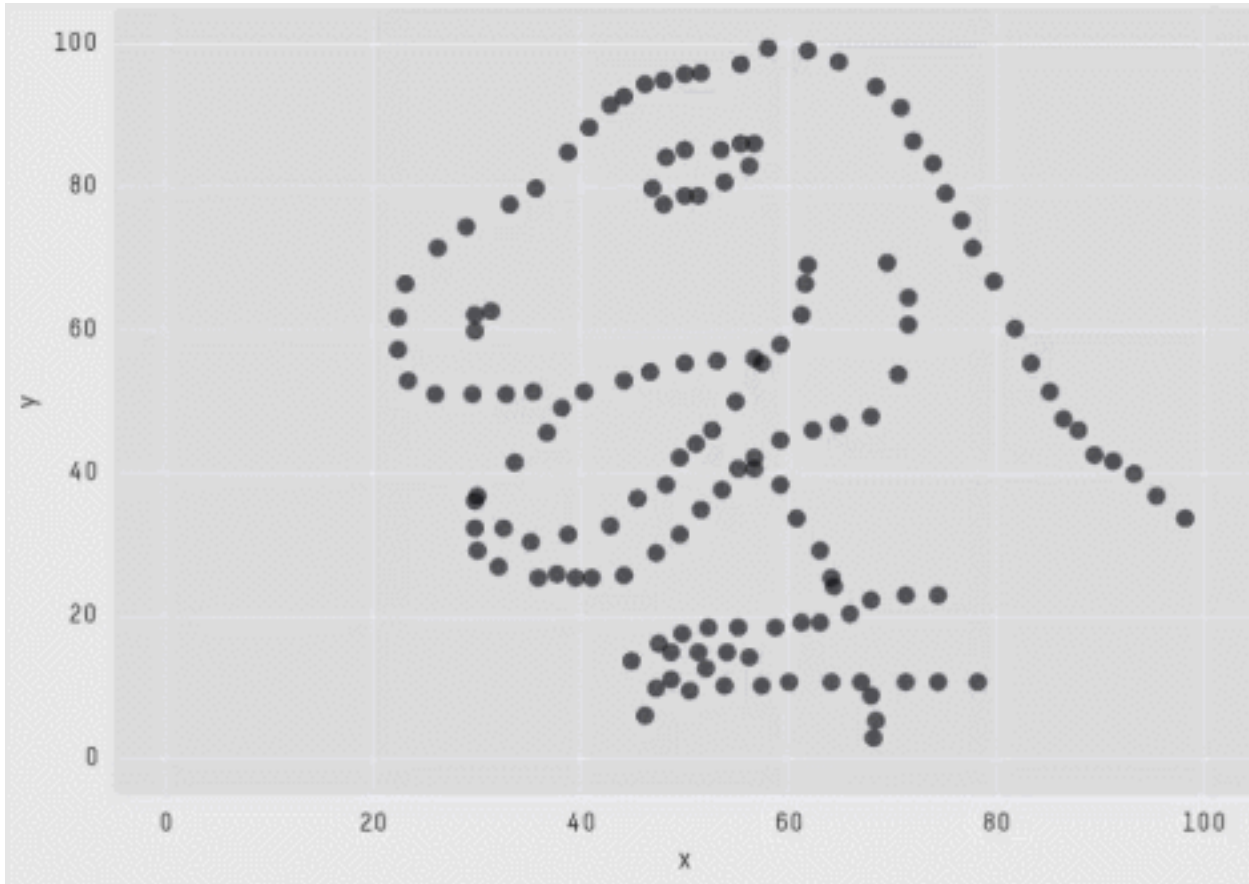
Just show me the data!

.pull-left-3.small-code[

```
head(my_data, 10)
```

```
## [38;5;246m# A tibble: 10 × 2 [39m
##           x           y
## [3m [38;5;246m<dbl> [39m [23m [3m [38;5;246m<dbl> [39m [23m
## [38;5;250m 1 [39m 55.4 97.2
## [38;5;250m 2 [39m 51.5 96.0
## [38;5;250m 3 [39m 46.2 94.5
## [38;5;250m 4 [39m 42.8 91.4
## [38;5;250m 5 [39m 40.8 88.3
```


oh no

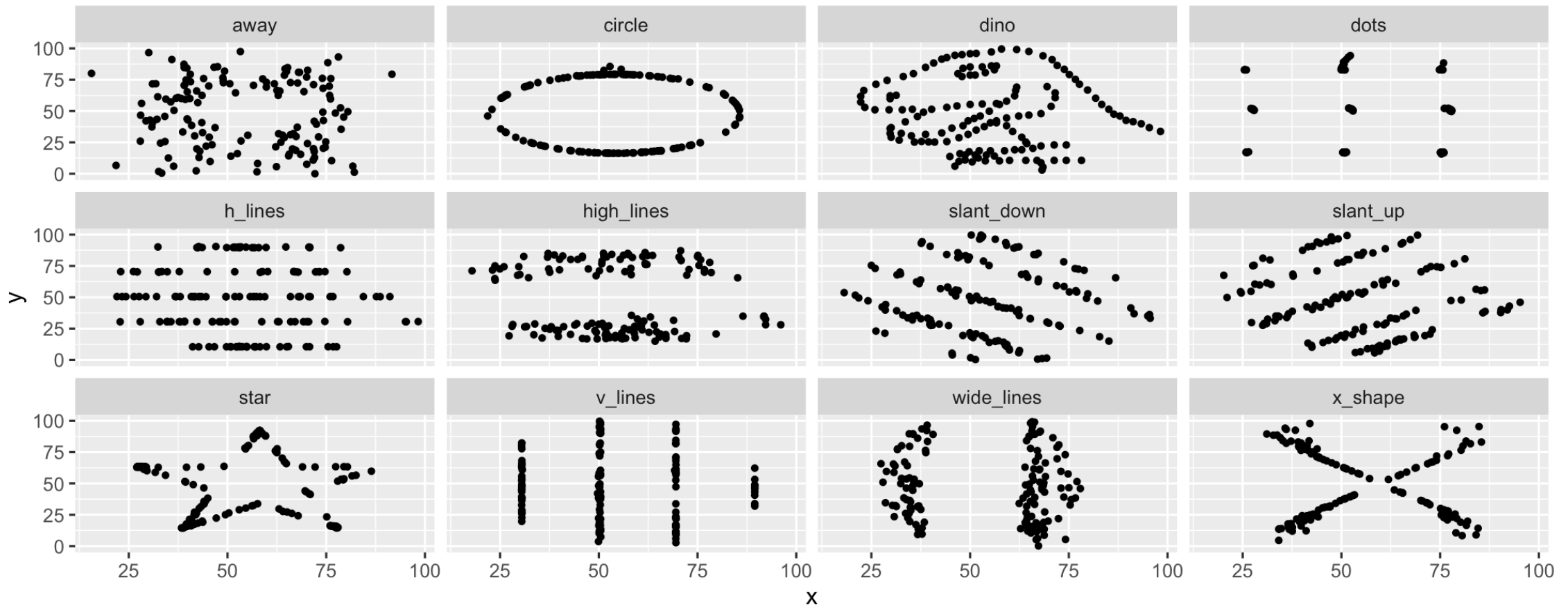


X Mean: 54.2659224
Y Mean: 47.8313999
X SD : 16.7649829
Y SD : 26.9342120
Corr. : -0.0642526

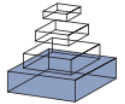
The Datasaurus Dozen

Raw data is not enough

Each of these has the same mean, standard deviation, variance, and correlation



Humans love patterns



Superior pattern processing is the essence of the evolved human brain

Mark P. Mattson^{1,2*}

¹ Laboratory of Neurosciences, National Institute on Aging Intramural Research Program, Baltimore, MD, USA

² Department of Neuroscience, Johns Hopkins University School of Medicine, Baltimore, MD, USA

Edited by:

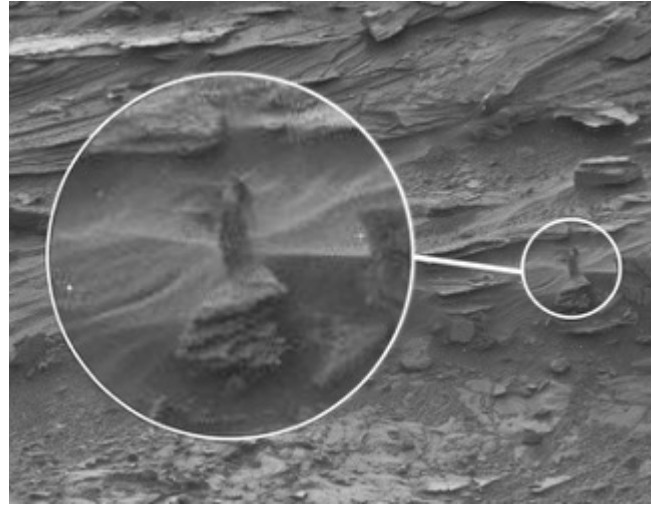
J. Michael Williams, Drexel
University, USA

Humans have long pondered the nature of their mind/brain and, particularly why its capacities for reasoning, communication and abstract thought are far superior to other species, including closely related anthropoids. This article considers superior pattern

<https://doi.org/10.3389/fnins.2014.00265>

(Sometimes we love them too much)

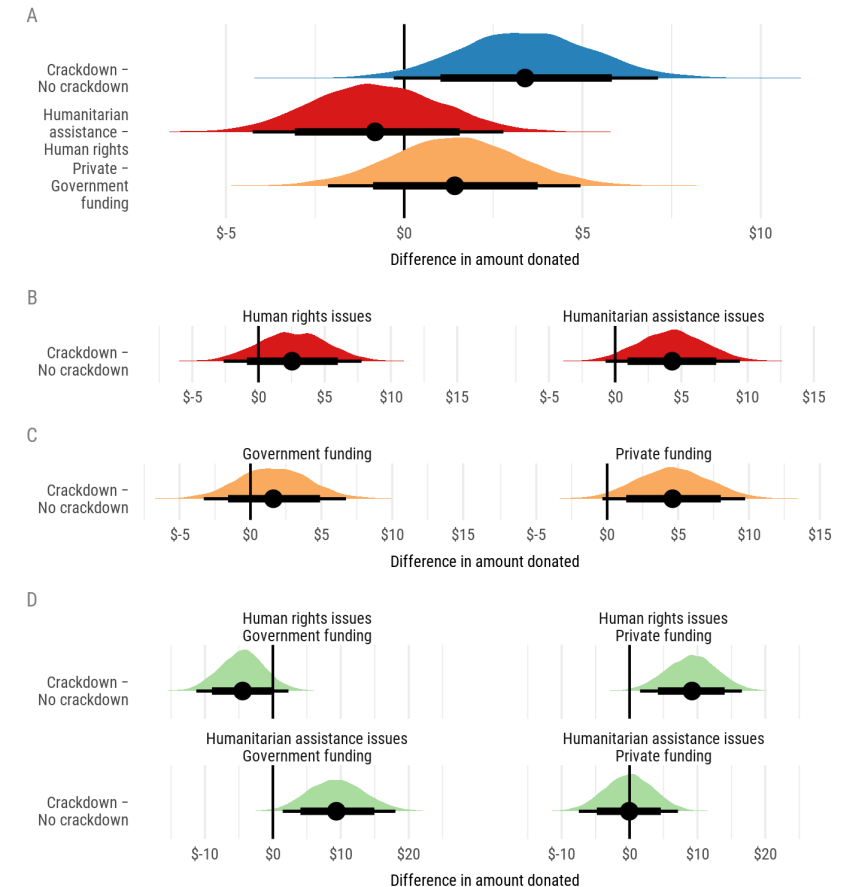
Pareidolia: seeing patterns that aren't there.



Beauty is necessary to see patterns

Table 2: Mean values and differences in means for amount donated in “crackdown” (treatment) and “no crackdown” (control) conditions; values represent posterior medians

H _{1b}	Amount _{Treatment}	Amount _{Control}	Δ	% Δ	$p(\Delta \neq 0)$
Crackdown – No crackdown	16.34	12.93	3.39	26.3%	0.97
Humanitarian assistance – Human rights	14.06	14.85	-0.82	-5.5%	0.67
Private – Government funding	15.13	13.71	1.42	10.4%	0.79
H _{2b} and H _{3b}	Amount _{Crackdown}	Amount _{No crackdown}	Δ	% Δ	$p(\Delta \neq 0)$
Human rights issues	17.4	14.86	2.54	17.2%	0.83
Humanitarian assistance issues	15.91	11.68	4.3	36.9%	0.95
Government funding	13.83	12.24	1.61	13.1%	0.74
Private funding	18.95	14.23	4.62	32.4%	0.97
H _{2b} and H _{3b} (nested)	Amount _{Crackdown}	Amount _{No crackdown}	Δ	% Δ	$p(\Delta \neq 0)$
Human rights issues, Government funding	10.56	15.15	-4.46	-29.5%	0.91
Human rights issues, Private funding	23.76	14.5	9.19	63.8%	0.99
Humanitarian assistance issues, Government funding	21.42	11.89	9.35	77.9%	0.99
Humanitarian assistance issues, Private funding	15.69	15.72	-0.05	-0.3%	0.51



Point shows posterior median; thick black lines show 80% credible interval; thin black lines show 95% credible interval

Beautiful visualizations

What makes a great visualization?

Truthful

Functional

Beautiful

Insightful

Enlightening

Alberto Cairo, *The Truthful Art*

What makes a great visualization?

"Graphical excellence is the well-designed presentation of interesting data—a matter of substance, of statistics, and of design ... [It] consists of complex ideas communicated with clarity, precision, and efficiency. ... [It] is that which gives to the viewer the greatest number of ideas in the shortest time with the least ink in the smallest space ... [It] is nearly always multivariate ... And graphical excellence requires telling the truth about the data."

Edward Tufte, *The Visual Display of Quantitative Information*, p. 51

What makes a great visualization?

Good aesthetics

No substantive issues

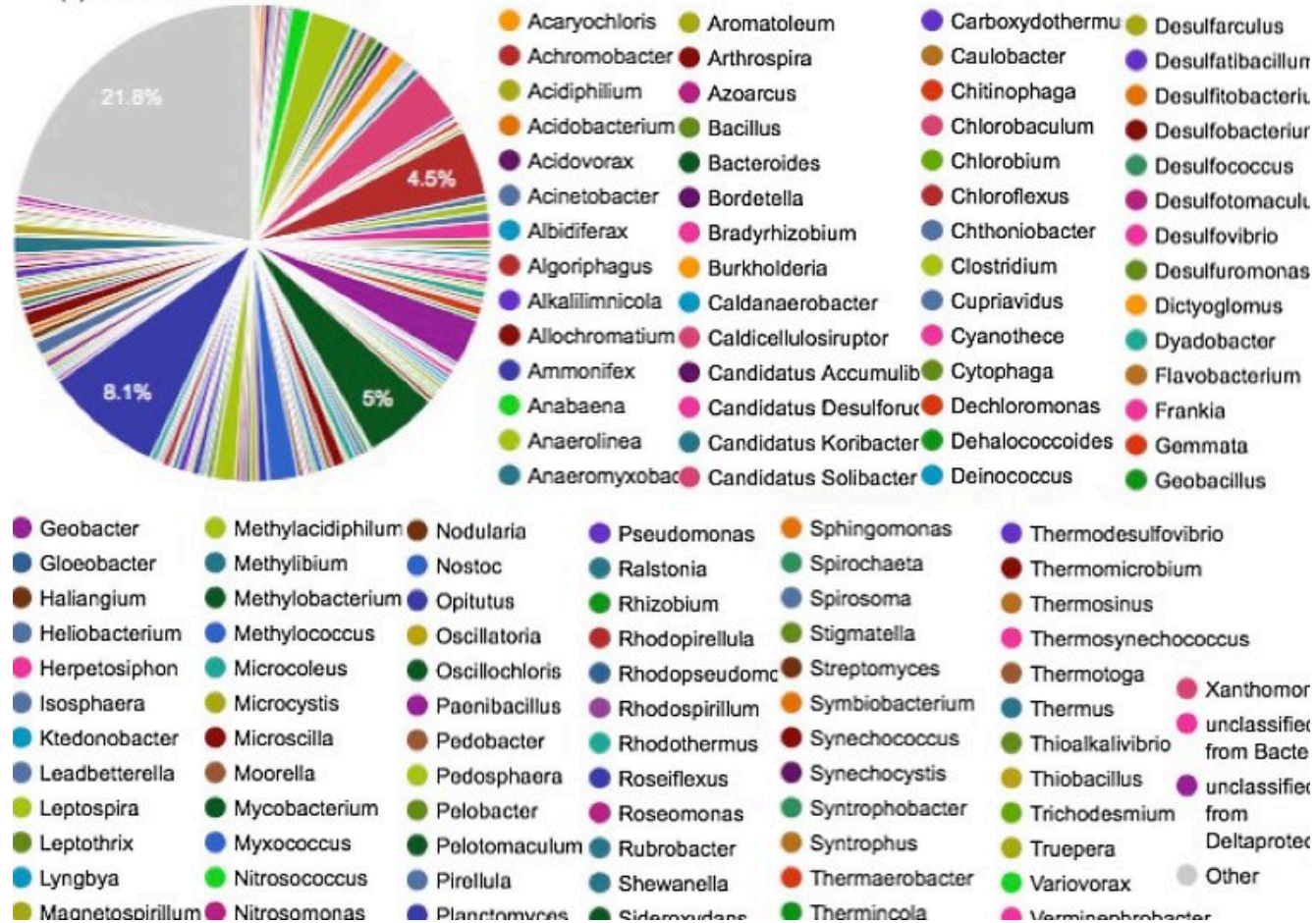
No perceptual issues

Honesty + good judgment

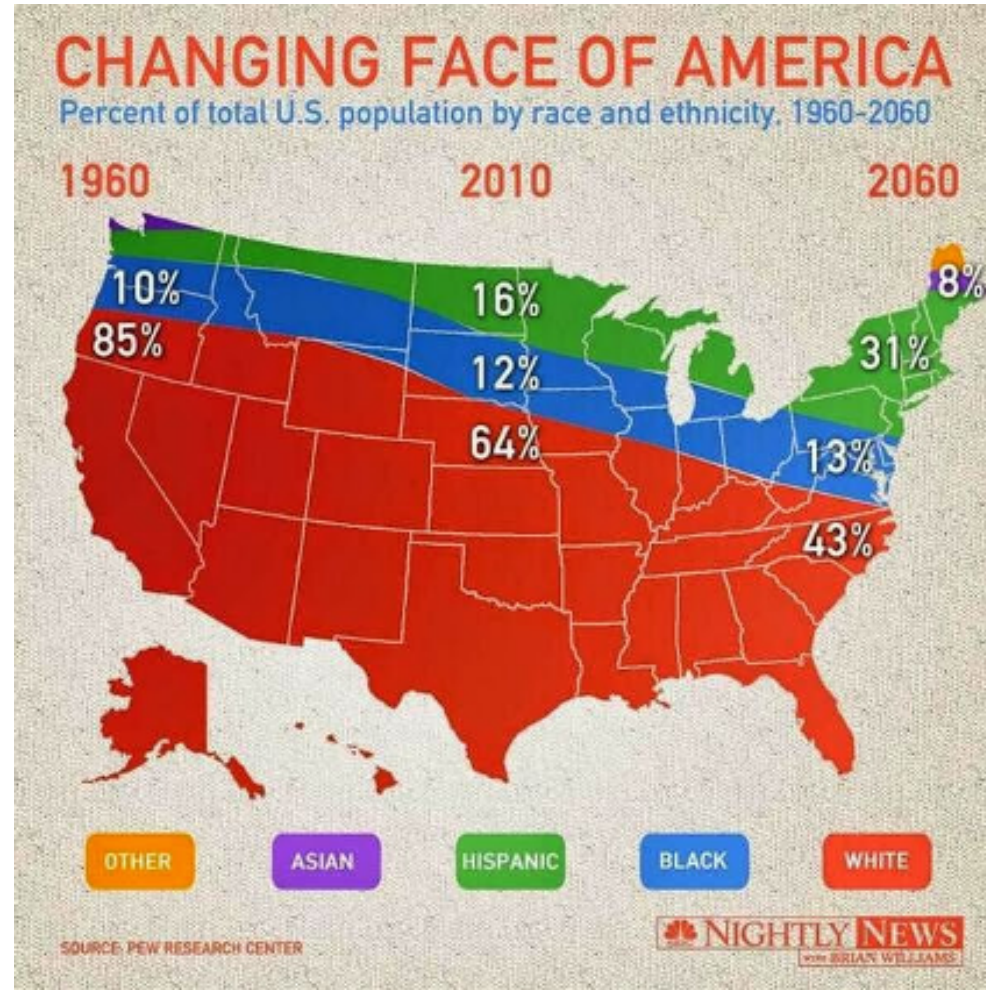
Kieran Healy, *Data Visualization: A Practical Introduction*

What's wrong?

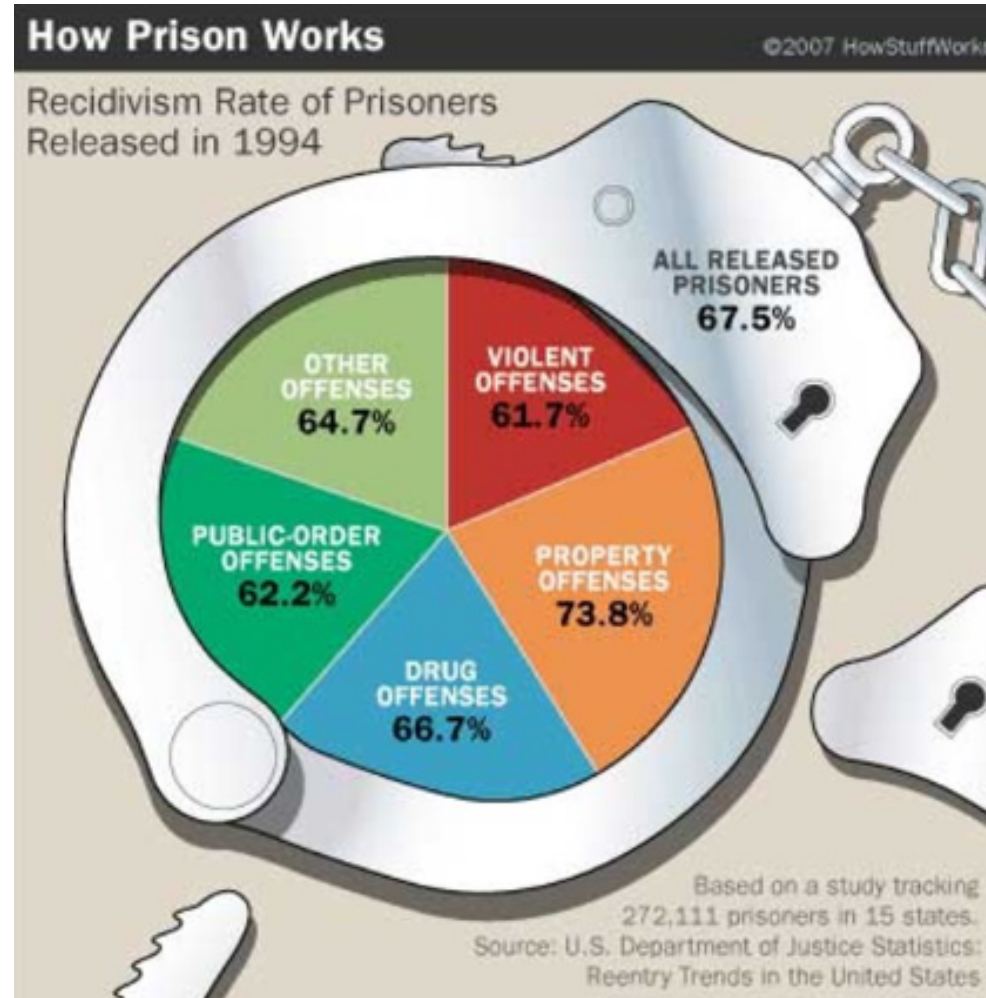
(f) Distribution of Genus



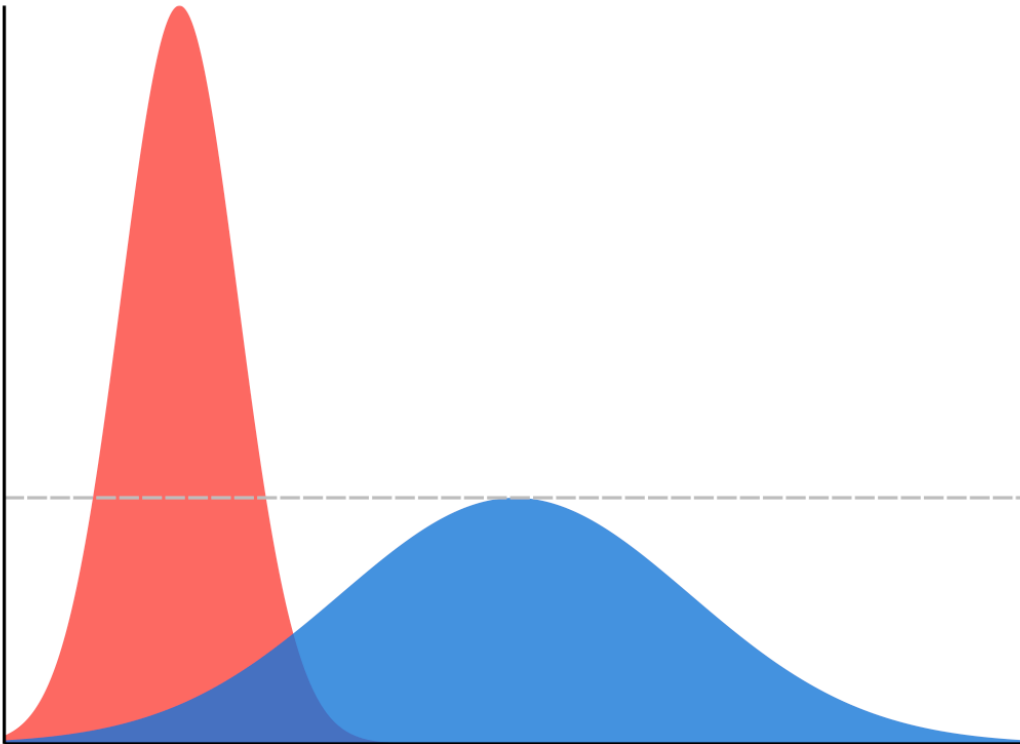
What's wrong?



What's wrong?



What's right?



Carl T. Bergstrom @CT_Bergstrom · Mar 6

3. There is a lot of complicated epidemiological modeling behind this idea, but this graphic strips all of that away, and discards irrelevant details to provide a straightforward story that people find easy to grasp at a glance.

It *simplifies* and *highlights* what matters.

6 198 1.8K

[Show replies](#)

Carl T. Bergstrom @CT_Bergstrom · Mar 6

4. I've seldom seen a piece of sci-comm matter so much. We have an opportunity to flatten the [#COVID19](#) [#coronavirus](#) epidemic curve by aggressive social distancing and other measures.

But people don't understand what the point is, if the virus is going to circulate broadly.

8 313 2K

[Show replies](#)

Carl T. Bergstrom @CT_Bergstrom · Mar 6

5. This graph provides the answer, powerfully and concisely.

And because of that, it has exploded across twitter and other media. I've used it myself a number of times. This graph is changing minds, and by changing minds, it is saving lives.

6 196 1.5K

Thread by Carl T. Bergstrom

Class details

Goal for the class

**Recognize and create
beautiful and truthful
visualizations with real
world data**

Plan for the class

Foundations

Truth and beauty
Graphic design principles
Mapping data to graphics



Core types of graphics

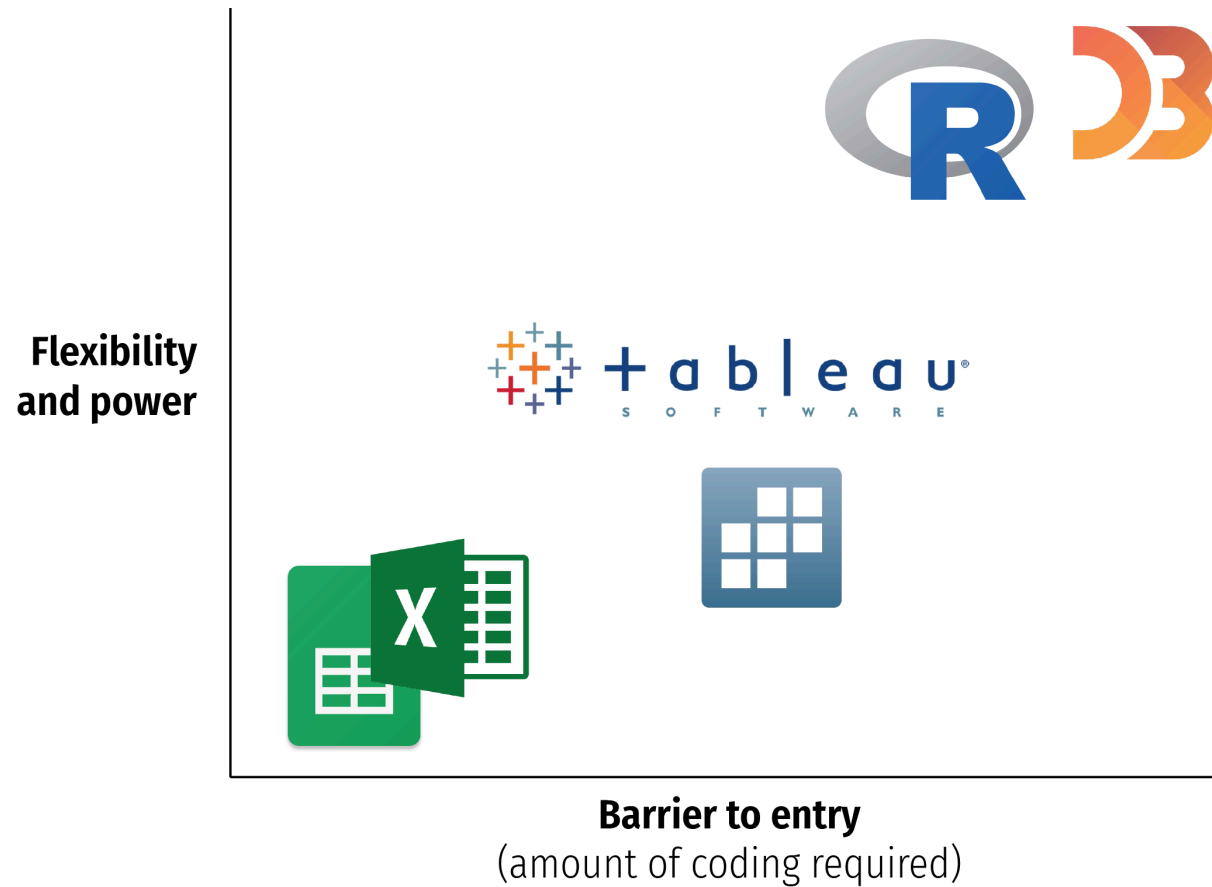
Amounts Proportions
Uncertainty Relationships
Comparisons Annotations



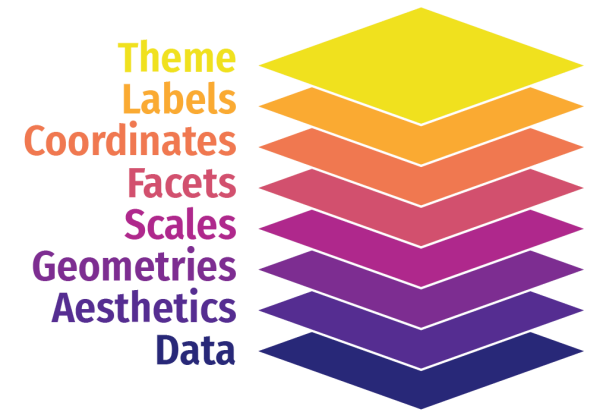
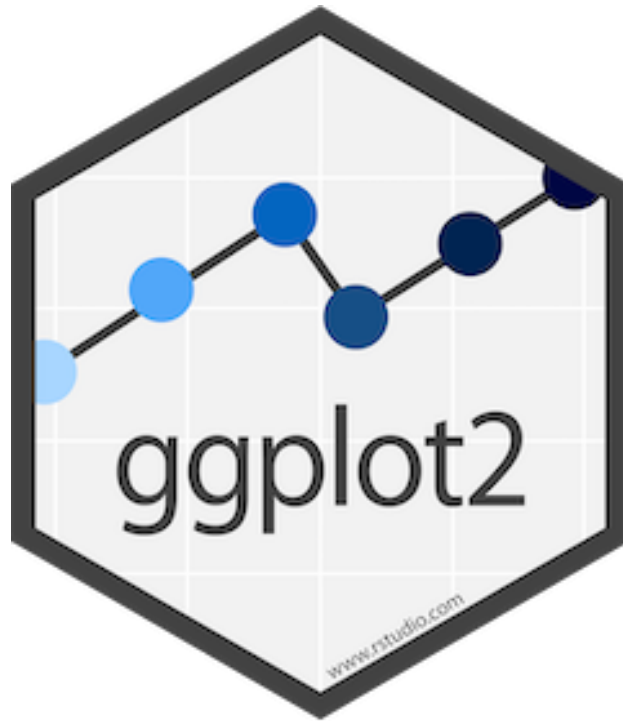
Special applications

Interactivity
Time Space Text
Enhancing graphics

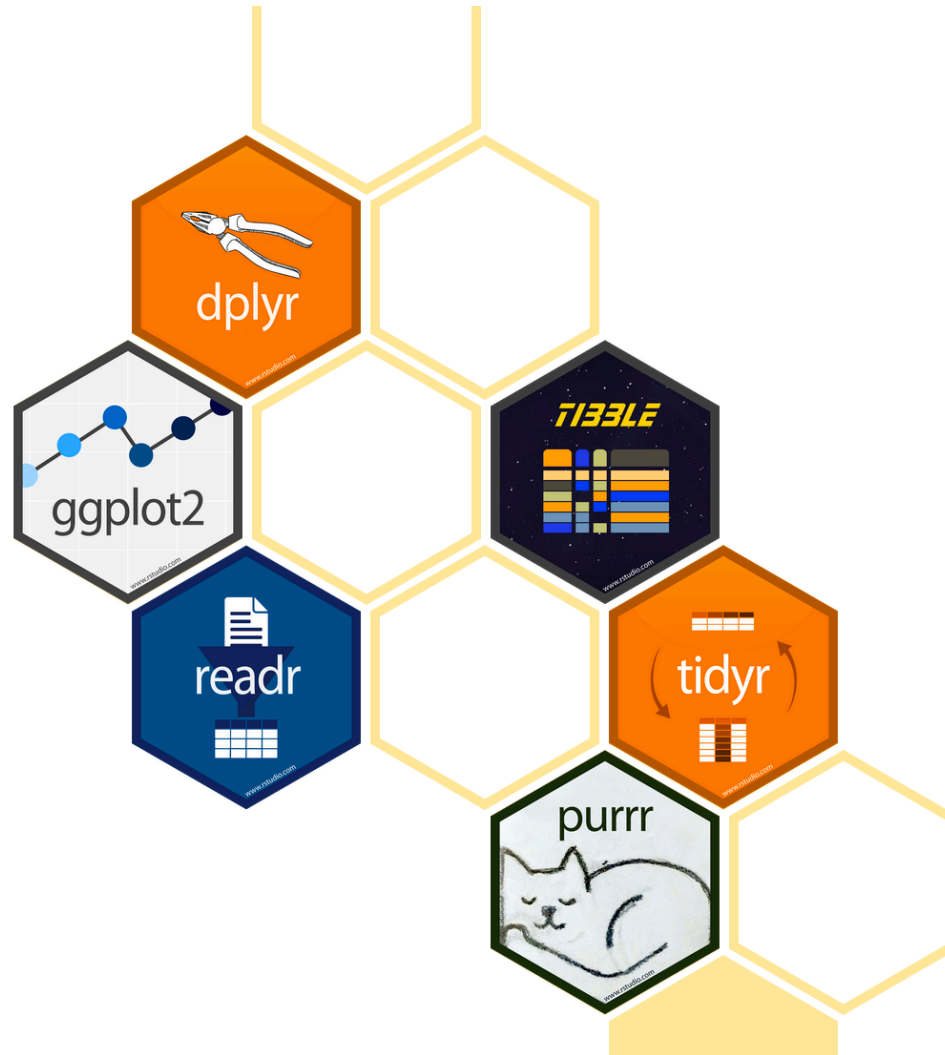
Class technology



Class technology



The tidyverse



Sucking

“There is no way of knowing nothing about a subject to knowing something about a subject without going through a period of much frustration and suckiness.”

“Push through. You’ll suck less.”

Hadley Wickham, author of `{ggplot2}`

Sucking



The New York Times

Opinion

SPORTING

(It's Great to) Suck at Something

By Karen Rinaldi

April 28, 2017



Karen Rinaldi, "(It's Great to) Suck at Something"